

A brief history of the evolution of mining is needed so that the innovative investor can better understand the current state of affairs for bitcoin and other cryptoassets. From there, it is easier to decide if this avenue of acquisition is appropriate. Even for those who have no interest in mining themselves, it's valuable to have a deeper understanding because for many cryptoassets mining is the means of new supply issuance and the security system underpinning transactions.

When Bitcoin's network was launched in January 2009, mining was the only method of acquiring bitcoin, and Satoshi Nakamoto and Hal Finney were the two main miners.² As we've discussed, new bitcoin is minted through the process of verifying and confirming transactions in Bitcoin's blockchain, the orchestration of which is a large part of the software that Satoshi created. In this way, it ensures the decentralized creation of the currency in controlled amounts, which prior to bitcoin had not been accomplished on a global scale.

The mining process for bitcoin is a continual cycle of hashing a few pieces of data together in pursuit of an output that meets a predetermined difficulty level, mainly the number of 0s that the output starts with. We call this output the *golden hash*. Recall that a hash function takes data—for example the text in this sentence—and hashes it into a fixed-length string of alphanumeric digits. While the output of a hash function is always of fixed length, the characters within it are unpredictable, and therefore changing one piece of data in the input can drastically change the output. It's called a golden hash because it bestows the privilege of that miner's block of